

DISTRIBUTION SUBSTATIONS WITH EDGE GENERATION

Design Basis: DB-Layer Postulated Events and Acceptance Criteria

**DBA-ES-DI-FC5: Distribution Facility Class with Edge
Generation**

A Child Document Under

DBA-ES: Electric Sector Design Basis Framework (v0.6)

DBA-ES FC5 and CntlCo Gap Analysis (v0.1) as Predecessor Document

Version 0.1 — DRAFT

DBA Consequence Envelope and MA Extension Deferred to v0.2

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Tim Roxey

Eclectic Technologies

For review by DisCo engineers, distribution planners, microgrid operators, defense installation energy managers, NERC reliability analysts, and state regulatory staff.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. DB-layer instrument only; DBA consequence envelope and MA extension deferred to v0.2. Inherits postulated event architecture from DBA-ES FC5 and CntlCo Gap Analysis (v0.1) as predecessor document. Established facility class definition (HV/EHV input; 69–110 kV output; edge generation; islanding capability). Developed a full postulated event taxonomy, including all inherited sector-level categories (PE, SE, CE, XC) and the locally defined MT (Mode Transition) category. Incorporated PE-SS (Station Service Loss) with three sub-cases as a series-significant postulated event. Established the Command Broker requirement for intentional islanding (MT-1) as a Bright Line architectural requirement. Established a 72-hour minimum fuel reserve at the rated islanded load as a series standard for FC5 facilities. Acknowledged dual-DB architecture per DBA-ES §3.4. Registered DBA-GAS §8.3 and DBA-OIL §8.3 Bright Line naming correction as Open Item OI-1.

Section 1 — Purpose, Scope, and Version Status

1.1 Purpose and Version Scope

This document establishes the Design Basis (DB) layer for distribution substations with edge generation capability within the Electric Sector Design Basis Series. It is the fifth facility-class instrument in the series, following DBA-ES-HY-FC1 (large hydroelectric storage), DBA-ES-TX-FC2 (large EHV transmission substations), DBA-ES-TX-FC3 (HVDC interconnects), and DBA-ES-GC-FC4 (natural gas compressor stations).

This is a DB-layer instrument. The DBA consequence envelope and the military action extension are deferred to v0.2. The rationale for this scoping decision is that the DB-layer architecture for FC5 was developed through a dedicated gap analysis instrument (DBA-ES FC5 and CntlCo Gap Analysis, v0.1, June 2026) that established the postulated event taxonomy and identified two architectural decisions requiring resolution before the full instrument could be drafted. Those decisions — the Command Broker requirement for intentional islanding and the minimum fuel reserve design basis — are resolved and incorporated in this document. The DBA and MA extension layers will be built on this foundation in v0.2.

The document establishes two series-significant architectural requirements that have implications beyond FC5. First, the Command Broker requirement for intentional islanding (MT-1 and MT-3) establishes that mode transitions with consequence potential beyond the facility boundary belong above the Bright Line. Second, the 72-hour minimum fuel reserve design basis establishes a series-standard emergency duration for islanded generation assets.

1.2 Facility Class Definition

A facility within the scope of DBA-ES-DI-FC5 must satisfy all three of the following criteria.

Physical tier. The substation receives power at HV or EHV (typically 115 kV and above) from the transmission system. It delivers power at distribution voltage (69 kV to 110 kV, or lower secondary distribution voltage) to downstream loads.

Edge generation or microgrid capability. The facility has one or more generation or storage assets connected to the distribution bus that can supply power to loads behind the substation when the transmission feed is unavailable. Under the lumped-parameter treatment, the entire collection of edge generation and storage assets within the islanding boundary is analyzed as a single generation resource. Generation assets may include synchronous generators, inverter-based resources (solar, wind, battery storage), fuel cells, or any combination thereof.

Islanding capability with Command Broker architecture. The facility's protection and control architecture supports intentional islanding — the ability to disconnect from the transmission feed and operate in a self-contained mode. The intentional islanding function must be implemented with a Command Broker authorization requirement as specified in Section 3.3. Anti-islanding protection as required by IEEE 1547 must be coordinated with the intentional islanding scheme through a transfer trip, permissive, or equivalent scheme that distinguishes authorized intentional islands from unauthorized unintended islands.

1.3 Predecessor Document

The DBA-ES FC5 and CntlCo Gap Analysis (v0.1, June 2026) is the predecessor document for this instrument. It established the facility class architecture, the lumped-parameter treatment rationale, the defense-critical load dependency anchor, and the preliminary postulated event taxonomy that this document develops into a full DB-layer instrument. All postulated event categories and events in this document are traceable to the gap analysis. Where this document differs from the gap analysis, this document governs.

1.4 Position in Hierarchy

Document Number: DBA-ES-DI-FC5-001

Series: Electric Sector Design Basis Series — Facility Class Instruments

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.6)

Predecessor Document: DBA-ES FC5 and CntlCo Gap Analysis (v0.1)

Section 2 — Facility Profile: Distribution Substation with Edge Generation

2.1 Physical Configuration

An FC5 facility is a distribution substation receiving transmission-level voltage through one or more HV or EHV supply transformers, with a distribution bus operating at 69 kV to 110 kV (or lower), and one or more generation or storage assets connected to the distribution bus or to a secondary bus fed from it. The generation and storage assets, their interconnection equipment, and the protection and control systems governing their operation and the islanding function are all within the FC5 entity boundary under the lumped-parameter treatment.

The control building houses the substation automation systems, protective relay panels, SCADA remote terminal unit, microgrid controller, UPS, and communications equipment. Station service power for these systems is the critical dependency that PE-SS addresses: all control and protection functions depend on continuous station service power, and sub-case (c) of PE-SS — simultaneous loss of station service and microgrid controller from a single initiating event — is the governing design basis concern specific to FC5.

2.2 The Lumped-Parameter Treatment

The lumped-parameter treatment is the analytical foundation of the FC5 facility class. The distribution substation, its protection and control systems, its station service infrastructure, and all edge generation and storage assets within the islanding boundary are analyzed as a single design basis entity. Individual component failures within this entity boundary are within-entity events (SE-2); the entity boundary is the analytical unit for consequence analysis.

The lumped-parameter treatment is analogous to the nuclear sector's treatment of a reactor and its associated safety systems as a single design basis entity, even though the components are physically distributed across the site. The analytical value is that it allows the design basis to address the compound failure modes — simultaneous loss of station service and microgrid controller (PE-SS sub-case (c)); simultaneous damage to substation and edge generation from a compound natural hazard (SE-3) — that are most consequential for the facility's resilience function.

2.3 Defense-Critical Load Dependency

The Defense Science Board report “More Fight — Less Fuel” (February 2008) found that 31 of 34 of the DoD's most critical facilities depend on commercial distribution infrastructure they do not control. This finding is the evidentiary anchor for the XC-1 postulated event. It establishes that the design basis for FC5 facilities serving defense-critical loads must treat the defense-critical consequence as a required analytical element.

The 72-hour minimum fuel reserve design basis (Section 3.4) is specifically calibrated to defense-critical load dependency: it provides a minimum resilience duration consistent with emergency response mobilization timelines, while acknowledging that defense-critical facilities with longer mission requirements must specify fuel reserves beyond the series minimum.

2.4 Regulatory Context

FC5 facilities span multiple regulatory jurisdictions. The HV/EHV supply transformer and the transmission interconnection are subject to NERC reliability standards and FERC jurisdiction. The distribution bus and downstream distribution assets are subject to state public utility commission jurisdiction. Edge generation assets interconnecting under IEEE 1547 are subject to state interconnection requirements and applicable utility tariffs. There is no federal regulatory standard that addresses the integrated FC5 entity — the combination of distribution substation, microgrid, islanding capability, and defense-critical load dependency — as a single design basis subject. This document supplies the design basis.

Section 3 — Series-Significant Architectural Requirements

3.1 Scope of This Section

Two architectural requirements established in this instrument have implications beyond FC5. They are presented in a dedicated section before the postulated event taxonomy to ensure their rationale is clearly stated and traceable to the series-level principles from which they derive.

3.2 The Bright Line Applied to Mode Transitions

The Bright Line, established as a series standard in DBA-ES §3.4 and DBA-DC-L1 §3.5, defines the boundary between autonomous ML system operation (below the Bright Line) and human Command Broker oversight (above it). The Bright Line applies to operational decisions, not only to information processing tasks. A mode transition — the deliberate change of an FC5 facility's operating state from grid-connected to islanded or from islanded to grid-connected — is an operational decision with consequence potential beyond the facility boundary (XC-3: reverse cascade from failed resynchronization) and is therefore a decision that belongs above the Bright Line.

The asymmetry between MT-1 (planned, above Bright Line) and MT-2 (automatic, below Bright Line) is deliberate and architecturally important. MT-2 is a deterministic protective response to a defined triggering condition (transmission fault, feed loss meeting automatic islanding criteria). It is formally testable: given the triggering condition, the protection system produces a defined output. It is therefore appropriate for autonomous operation below the Bright Line. A defined physical condition does not trigger MT-1; it is initiated by an operator's decision to enter islanded mode. The decision involves assessing current operating state, edge generation readiness, load profile, and communications status — a non-deterministic judgment that requires human oversight. MT-3 requires the same judgment in reverse.

3.3 Command Broker Requirement for Intentional Islanding

Intentional islanding (MT-1) and resynchronization (MT-3) require human Command Broker authorization as a precondition for execution. The Command Broker requirement is implemented as follows.

Authorization channel. The Command Broker must be reachable via a communications channel that is physically independent of the primary SCADA path. The authorization channel must remain functional under the conditions that typically precede an MT-1 decision — transmission fault, grid instability, or planned maintenance switching — which may degrade or eliminate the primary SCADA communications path. The specific design basis for the authorization channel is deferred to DI-5.

Prerequisite checklist. The MT-1 authorization protocol must include verification of: (a) station service continuity through the transition (PE-SS sub-case (c) prevention); (b) edge generation readiness and fuel reserve adequacy; (c) relay settings consistent with the target islanded operating mode (CE-3 prevention); and (d) load shedding plan for loads that cannot be sustained in islanded mode. The Command Broker must confirm each prerequisite before issuing the authorization command.

Automatic fallback for MT-2. The Command Broker authorization requirement applies only to intentional transitions. If the protection system automatically islands the facility under MT-2 conditions (transmission fault, unplanned feed loss), the Command Broker is not required and the automatic response must not be delayed awaiting authorization. The MT-2 automatic response operates below the Bright Line as a deterministic protective action.

3.4 Minimum Fuel Reserve Design Basis

The minimum fuel reserve design basis for FC5 is 72 continuous hours of operation at rated islanded load for all edge generation assets combined, without resupply. This is a series standard established at this instrument version.

The 72-hour standard is calibrated to three reference points: the 72-hour station service continuity criterion established in DBA-ES-TX-FC2; the 72-hour backup power posture threshold used in Tier 3 acceptance criteria across the DBA-ES series; and emergency response mobilization timelines that establish 72 hours as the minimum period before organized fuel resupply operations can be expected to reach an FC5 facility under adverse conditions.

The 72-hour standard is a minimum. FC5 facilities serving defense-critical loads with mission requirements extending beyond 72 hours must specify fuel reserves consistent with the mission requirement, not only the series minimum. The service area critical load inventory (DI-4) identifies the binding mission requirement duration for each FC5 installation.

Fuel reserves must be physically present on-site and must not be contingent on delivery during an emergency. For diesel or propane generation, this means on-site tank capacity. For gas-fired edge generation drawing from a distribution gas supply, the gas supply design basis under

islanded conditions must be established (DI-3); if the gas supply is not available under islanded conditions, on-site compressed gas or LNG backup must be provided.

Section 4 — Design Basis: FC5 Postulated Event Categories

4.1 Sector-Level Inheritance and Locally Defined Category

FC5 inherits four sector-level postulated event categories from DBA-ES §3.5: PE (Power Supply and Grid Interface), SE (Structural and Physical Integrity), CE (Control and Communications), and XC (Cross-Sector Cascade). A fifth category — MT (Mode Transition Events) — is locally defined for FC5, following the precedent of the DBA-DC series' WM (Workload Migration) category. MT events are unique to the islanding-capable architecture of the FC5 facility class and have no analog in any other DBA-ES facility class instrument.

4.2 Category PE — Power Supply and Grid Interface Events

PE events at FC5 run in both directions: upstream (the substation receives power from the transmission system) and downstream (the edge generation assets supply power to the islanded distribution bus). PE-SS is identified as a series-significant postulated event because sub-case (c) — simultaneous loss of station service and microgrid controller from a single initiating event — is a correlated failure mode unique to FC5 with no analog in the transmission-tier FC instruments.

Code	Event	Initiating Condition	Governing Concern
PE-1	Short-duration transmission feed loss (≤ 15 minutes)	Loss of HV/EHV transmission supply to the distribution substation for up to fifteen minutes. UPS and onsite generation bridge the station service gap. The facility remains in grid-connected mode or transitions to islanded operation if the automatic protection scheme permits.	The transmission feed loss must not interrupt station service. Station service continuity through PE-1 is a mandatory acceptance criterion: the substation's control, protection, and communications systems must remain energized throughout the event and through any mode transition it initiates.
PE-2	Extended transmission feed loss (15 minutes to 72 hours)	Loss of HV/EHV transmission supply beyond fifteen minutes, requiring sustained islanded operation from edge generation or microgrid assets. The facility operates as a self-contained island, with the edge generation assets providing the sole supply to loads behind the islanding boundary.	Sustained islanded operation requires adequate edge generation capacity and fuel supply. The 72-hour fuel reserve design basis (Section 3.4) bounds the PE-2 design basis duration. Generation dispatch within the island must maintain frequency and voltage within the ride-through thresholds of the local protection systems.

PE-3	Extended transmission feed loss (beyond 72 hours)	Loss of HV/EHV transmission supply beyond 72 hours, beyond the minimum fuel reserve design basis duration. Edge generation fuel may be exhausted. The facility must transition to a minimum viable operating state: shedding non-essential loads while maintaining power to the highest-priority loads behind the islanding boundary.	PE-3 is the bounding PE event for FC5. The design basis must define the minimum viable operating state: which loads are shed in which sequence, what the minimum fuel reserve provides, and at what point the facility operator must seek emergency fuel resupply or accept full shutdown.
PE-SS	Station service (hotel load) loss	Loss of AC station service power to the control shack, relay panels, SCADA, communications equipment, and protection systems. Three sub-cases govern the analysis: (a) Transmission feed lost, UPS intact: station service maintained for UPS design duration. Normal mode transition to islanded operation may proceed with Command Broker authorization. (b) Transmission feed lost, UPS failed or depleted: immediate loss of station service. Substation goes operationally blind. (c) Transmission feed is lost during islanded operation, where the microgrid architecture does not provide an independent station service supply: simultaneous loss of the substation control function and the microgrid controller from the same initiating event.	Sub-case (c) is the governing concern for FC5 and a series-significant postulated event. If the microgrid controller and the substation protection systems share the same station service bus that fails during the islanding transition, the Command Broker loses the ability to authorize MT-1, and the automatic protection systems lose the ability to execute MT-2 at the same moment. Station service continuity through any mode transition is a mandatory acceptance criterion. Sub-case (c) must be addressed through an independent station service supply sourced from the microgrid secondary bus, not from the transmission-side station service transformer.
PE-4	Voltage excursion during islanded operation	Sustained voltage outside the ANSI C84.1 Range A at the distribution bus during islanded operation, caused by generation-load imbalance within the island. Inverter-based resources may not provide adequate reactive power support under all islanded conditions.	Voltage regulation capability of islanded inverters and synchronous generators must be demonstrated for the full range of load conditions within the islanding boundary. Under-voltage ride-through capability of load equipment connected to the islanded bus is a protection coordination requirement.
PE-5	Frequency excursion during islanded operation	Frequency deviation beyond PRC-024 ride-through thresholds during islanded operation due to generation-load imbalance within the island. The islanded system has no external frequency reference or synchronous inertial support from the transmission grid.	Governor response adequacy of islanded generation assets is the governing concern. Inverter-based resources require a synthetic frequency response; the design basis must specify the frequency response capability required for each generation technology within the islanding boundary.

4.3 Category MT — Mode Transition Events (Locally Defined)

MT events address the transitions between grid-connected and islanded operating modes, and the failure mode that leaves the facility in an indeterminate state. The Command Broker requirement for MT-1 and MT-3 is a binding architectural requirement established in Section 3.3; it applies to every MT-1 and MT-3 event without exception.

Code	Event	Initiating Condition	Governing Concern
MT-1	Planned grid-to-island transition (Command Broker authorized)	Deliberate operator-commanded transition from grid-connected to islanded operation, authorized by a human Command Broker above the Bright Line. The Command Broker reviews the current operating state — transmission feed status, edge generation readiness, load profile, station service status — and authorizes the islanding command. The automatic protection system executes the transition on receipt of the authorized command.	Command Broker authorization is a mandatory prerequisite for MT-1. The transition may not be executed autonomously by any ML or Gen AI system without Command Broker authorization. The Command Broker must be reachable — via a communications channel independent of the primary SCADA path — before the MT-1 command can be issued. Station service continuity through the MT-1 transition is a mandatory acceptance criterion.
MT-2	Unplanned (forced) grid-to-island transition (automatic)	Automatic islanding triggered by grid fault, transmission feed loss, or protective relay operation, without deliberate operator command. The automatic protection system executes the islanding transition below the Bright Line as a deterministic protective response. No Command Broker authorization is required for MT-2.	MT-2 is the anti-islanding resolution: the protection scheme must distinguish between an unintended island — which anti-islanding protection would prevent — and an authorized intentional island. MT-2 is the case where islanding is permitted automatically because the triggering condition (transmission fault, feed loss) satisfies the protection scheme's automatic islanding criteria. Automatic generation control and load shedding within the island must stabilize frequency and voltage within the ride-through window without Command Broker involvement.
MT-3	Island-to-grid resynchronization	Restoration of the grid connection following an islanding event. Requires frequency, voltage, and phase angle matching at the interconnection point within the tolerance of the synchronizing relay or operator-controlled synchronizing scheme. Command Broker authorization is required for MT-	Out-of-phase reclosure during MT-3 — caused by insufficient synchronization accuracy — can damage generation equipment, protection systems, and connected loads, and may initiate a second unplanned islanding event. The synchronizing relay settings and

		3, consistent with the MT-1 requirement: resynchronization is a deliberate operational action that must not be executed autonomously.	the Command Broker authorization protocol must together bound the probability of out-of-phase reclosure to an acceptable level.
MT-4	Failed transition — indeterminate operating state	A transition attempt (in either direction) that does not complete: the interconnection breaker fails to open during MT-1 or MT-2, or fails to close during MT-3. The system is neither fully islanded nor fully grid-connected. Station service, protection system response, and generation dispatch are all in an undefined state relative to their design operating modes.	The failed transition state must be addressed in the protection coordination design as a specific postulated condition. The protection system must have defined responses for MT-4 in each direction that restore the facility to a known operating state — either fully islanded or fully grid-connected — without loss of station service and without cascading trip of generation assets.

4.4 Category SE — Structural and Physical Integrity Events

SE events at FC5 address both the distribution substation structure and the edge generation and storage assets within the lumped-parameter boundary. The compound natural hazard case (SE-3) is the bounding SE event: it simultaneously removes the transmission feed and the edge generation capacity, eliminating the resilience function that FC5 is designed to provide.

Code	Event	Initiating Condition	Governing Concern
SE-1	Distribution substation structural damage	Physical damage to the distribution substation structure, transformer banks, or switchgear from equipment failure, severe weather, or external physical event—Cause-agnostic at the DB layer. Large distribution transformers (69–110 kV) carry replacement lead times of six to eighteen months.	The RT_HVLLC concept applies at the distribution level with a shorter lead time range than at EHV. A distribution transformer requiring replacement produces a consequence duration that may exceed the 72-hour fuel reserve design basis of the edge generation assets, requiring interim supply arrangements.
SE-2	Edge generation or microgrid asset physical damage	Physical damage to generation units, storage systems, or interconnection equipment within the microgrid boundary. Under the lumped-parameter treatment, this is a within-entity event that reduces islanded generation capacity. Partial damage produces reduced islanded capacity; complete damage eliminates islanding capability.	The governing concern is partial loss of microgrid generation during islanded operation. The design basis must define the minimum viable islanded generation state — the minimum generation capacity that can sustain the highest-priority loads behind the islanding boundary — and demonstrate that at least this capacity remains available

			following the bounding SE-2 damage scenario.
SE-3	Natural hazard — compound event	Simultaneous or sequential occurrence of two natural hazard events affecting both the substation and the edge generation assets from the same initiating condition. The governing compound case for FC5 is simultaneous loss of the transmission feed (from the natural hazard's effect on the transmission system) and loss of edge generation capacity (from the same hazard's direct effect on the microgrid assets). Winter Storm Uri demonstrated this compound mode at scale: generation assets and transmission infrastructure failed simultaneously from the same weather event.	The compound case eliminates both supply paths simultaneously. With no transmission feed and reduced or eliminated edge generation, the facility cannot sustain islanded operation and faces complete load loss. This is the bounding SE case for FC5: it removes the resilience function that the microgrid was designed to provide at the same moment that resilience is most needed.

4.5 Category CE — Control and Communications Events

CE events at FC5 operate across three distinct control layers: the substation SCADA and protection system, the microgrid controller, and the Command Broker communications channel. Events affecting any one layer have mode-dependent consequences. CE-1, during islanded operation, blocks the Command Broker's ability to authorize MT-3, creating a potential isolation trap: the facility cannot resynchronize without Command Broker authorization, and the Command Broker cannot be reached because CE-1 has removed the SCADA communications path. The independent authorization channel required by the MT-1/MT-3 Command Broker requirement (Section 3.3) directly addresses this scenario.

Code	Event	Initiating Condition	Governing Concern
CE-1	Loss of substation SCADA and remote monitoring	Loss of communications between the distribution substation and the CntlCo (DisCo control center or ISO/RTO). The substation operates in local mode without remote visibility or command capability.	CE-1 in grid-connected mode is serious but bounded. CE-1 during islanded operation is more consequential: it removes the CntlCo's situational awareness of the island's operating state and complicates MT-3 resynchronization coordination. The Command Broker cannot be reached via the standard SCADA path if CE-1 has occurred; the MT-1 and MT-3 Command Broker authorization protocols must specify an independent communications channel.

CE-2	Loss of microgrid controller communications	Loss of communications between the microgrid controller and the individual generation or storage assets it coordinates. The controller loses the ability to dispatch generation, manage the state of charge, or execute load shedding commands within the island.	CE-2 during islanded operation is the most immediately destabilizing CE event for FC5. The microgrid controller is the frequency and voltage regulation authority within the island. Its loss under islanded conditions removes the primary control layer. The design basis must establish fallback local control modes for each generation asset that maintain minimum stability without microgrid controller coordination.
CE-3	Protective relay miscoordination across operating modes	Protective relay settings optimized for one operating mode produce an incorrect response in another. Relay settings must be valid across grid-connected, islanded, and transition operating modes. Mode-dependent relay settings require either adaptive relaying schemes or confirmation that settings valid for one mode do not produce harmful responses in other modes.	CE-3 is a design-basis gap, not an operational event: if relay settings are not mode-aware, the vulnerability is latent and present in all operating states. The MT-1 Command Broker authorization protocol should include verification that relay settings are consistent with the target islanded operating mode before the transition is authorized.

4.6 Category XC — Cross-Sector Cascade Events

XC events at FC5 operate in two directions: downward into the critical loads behind the islanding boundary (XC-1, XC-2) and upward into the transmission system through the resynchronization pathway (XC-3). The Command Broker requirement for MT-3 is partly motivated by XC-3: a correctly authorized and executed resynchronization eliminates the upward cascade risk, while an autonomous or improperly executed resynchronization introduces it.

Code	Event	Initiating Condition	Governing Concern
XC-1	Loss of resilience for defense-critical load	FC5 entity loss — substation, microgrid, or both — removes power from a defense-critical facility that depends on commercial distribution infrastructure it does not control—anchored on the Defense Science Board finding (2008) that 31 of 34 DoD most-critical assets depend on commercial DisCo infrastructure.	The XC-1 consequence is not the load loss itself but the degradation of a national defense capability. The design basis must identify which loads behind the FC5 entity carry XC-1 consequence potential. For those loads, the 72-hour fuel reserve design basis must be evaluated against the mission requirement duration of the dependent defense-critical facility.

XC-2	Loss of resilience infrastructure for dependent critical services	FC5 entity loss degrades the microgrid's function as the resilience resource for critical loads — hospitals, water treatment, emergency communications — that depend on the islanding capability for power continuity when the transmission grid is unavailable.	The microgrid is often the last-resort resilience resource for loads behind it. Its simultaneous loss with transmission grid unavailability eliminates all resilience margins at once. This is the compound consequence scenario that most directly demonstrates the value of the FC5 facility class: the design basis exists because the microgrid's resilience function must be analyzed and bounded, not simply assumed.
XC-3	Reverse cascade from failed resynchronization.	A failed or out-of-tolerance MT-3 resynchronization produces an out-of-phase reclosure that propagates a disturbance onto the transmission system, potentially triggering a protection response at the TransCo tier. This is the upward cascade pathway unique to FC5 in the DBA-ES series.	XC-3 establishes that a distribution-tier event can cascade upward into the transmission system through the resynchronization pathway. The Command Broker authorization requirement for MT-3 is partly motivated by XC-3: a correctly executed MT-3 eliminates the out-of-phase reclosure risk, while an autonomous or rushed resynchronization introduces it.

4.7 Dual Design Basis Architecture

Consistent with DBA-ES §3.4, an FC5 facility may simultaneously carry a sector-specific physical Design Basis (this document) and an AI Governance Design Basis under the AI Governance series. The Command Broker architecture established in Section 3.3 is itself a Bright Line governance structure applicable to the physical MT events; the AI Governance Design Basis extends this architecture to AI systems within the microgrid control layer.

Microgrid controller ML. The microgrid controller's dispatch optimization, state-of-charge management, and frequency regulation functions are ML systems operating below the Bright Line — deterministic, formally testable, autonomous operation permitted. However, the ML optimization must not override the minimum fuel reserve design basis or reduce linepack below the level required for the 72-hour design basis duration. The AI Governance Design Basis must establish minimum fuel reserve constraints that the ML optimization cannot breach.

Gen AI operational decision support. If Gen AI tools are used to support the Command Broker's MT-1 or MT-3 authorization decisions — for example, tools that synthesize grid status, edge generation readiness, and load profile into a recommendation — those tools require a separate Command Broker layer above the Bright Line. The Gen AI tool may inform, but may not replace the human Command Broker authorization. The independence requirement of the Influencer

Broker framework — the payload-generating system and the disclosure-generating system must be from different developers — applies to any Gen AI tools in the MT authorization workflow.

Section 5 — DB-Layer Acceptance Criteria

The following table establishes the DB-layer acceptance criteria for FC5. These criteria govern the design basis at the DB level; DBA-level acceptance criteria incorporating consequence analysis will be established in v0.2.

Criterion	Requirement	Source / Rationale
Station service continuity — all modes	Station service power must be maintained continuously through any PE event and any MT event. Sub-case (c) of PE-SS — simultaneous loss of station service and microgrid controller from a single initiating event — must be prevented by design through an independent station service supply sourced from the microgrid secondary bus.	Derived from PE-SS sub-case (c) as the governing FC5-specific concern. Consistent with station service continuity requirements in DBA-ES-TX-FC2 and DBA-ES-TX-FC3.
Command Broker for MT-1 and MT-3	Intentional islanding (MT-1) and resynchronization (MT-3) require human Command Broker authorization above the Bright Line before execution. The Command Broker must be reachable via a communications channel independent of the primary SCADA path. No ML or Gen AI system may execute MT-1 or MT-3 without Command Broker authorization.	Bright Line / Command Broker architecture per DBA-ES §3.4 and DBA-DC-L1 §3.5. MT-1 and MT-3 are non-deterministic operational decisions with consequence potential beyond the facility boundary (XC-3); they belong above the Bright Line.
Automatic MT-2 without Command Broker	Unplanned islanding (MT-2) must be executable automatically by the protection system below the Bright Line without Command Broker involvement. The protection scheme must distinguish MT-2 conditions (authorized automatic island) from anti-islanding trip conditions (unauthorized unintended island) through a transfer trip, permissive, or equivalent scheme.	Anti-islanding coordination requirement per IEEE 1547-2018. The Bright Line asymmetry — MT-1 above, MT-2 below — resolves the tension between anti-islanding protection and intentional islanding capability.
72-hour minimum fuel reserve	The FC5 facility must maintain a minimum fuel reserve sufficient to sustain all edge generation assets at rated islanded load for 72 continuous hours without resupply. Fuel reserves must be physically present on-site — not contingent on delivery during an emergency. For gas-fired edge generation, the 72-hour reserve must account for the fuel gas supply design basis under islanded conditions.	The series standard is established at this instrument version. Consistent with the 72-hour station service continuity criterion in DBA-ES-TX-FC2 and the 72-hour backup power posture threshold in Tier 3 acceptance criteria across the DBA-ES series.

Minimum viable islanded generation state	The design basis must define the minimum generation capacity that can sustain the highest-priority loads behind the islanding boundary under the bounding SE-2 damage scenario. This minimum viable state must be documented and must be achievable through load shedding without Command Broker involvement if the Commander Broker is unreachable.	Derived from the SE-2 bounding case and the XC-1 defense-critical load dependency. Consistent with the ASSC concept in the DBA-MA nuclear series, the minimum safe operating state must be achievable autonomously when external oversight is unavailable.
Mode-aware relay settings	Protective relay settings must be valid across grid-connected, islanded, and transition operating modes. Where single fixed settings cannot satisfy all modes, adaptive relaying or mode-switching schemes must be installed and demonstrated to be correct for each operating mode. CE-3 relay miscoordination must be explicitly addressed in the protection coordination study.	Derived from CE-3 as a design-basis gap. The MT-1 Command Broker authorization protocol must include relay setting verification as a prerequisite for transition authorization.

Section 6 — Open Items and Deferred Items

Item	Description	Resolution Criterion
OI-1	DBA-GAS §8.3 and DBA-OIL §8.3 Bright Line naming correction. Both documents use “Bright Line” to mean federal coordination triggers — a meaning distinct from the ML/AI governance concept established in DBA-DC-L1 and adopted as a series of standards in DBA-ES §3.4. Both documents must be revised to use a distinct term for federal coordination triggers (for example, “Federal Coordination Threshold”). All cross-references must be updated. Bright Line must be used exclusively in the ML/AI governance sense across all DBA series instruments.	DBA-GAS and DBA-OIL revised; Bright Line naming consistent across all DBA series documents.
DI-1	DBA Consequence Envelope (v0.2). The full DBA non-adversarial consequence envelope for FC5 is deferred to v0.2. This includes: bounding event identification for each postulated event category; non-adversarial consequence analysis at Tier 1, Tier 2, and Tier 3; extended consequence regime analysis for SE-1 (distribution transformer replacement); and the quantitative GSD-equivalent metric for FC5 (the duration of critical load supply interruption following FC5 entity loss).	DBA consequence envelope completed in DBA-ES-DI-FC5 v0.2.
DI-2	MA Extension (v0.2). The military action extension for FC5 is deferred to v0.2. This includes: MA extension applicability assessment against DBA-ES trigger criteria; application of the four MA characteristics (intentionality, persistence, precision, denial) to each postulated event category; governing MA scenarios including coordinated attack on FC5 entity and FC2 entities in the same service area; and acceptance criteria for the MA extension.	MA extension completed in DBA-ES-DI-FC5 v0.2.

DI-3	Fuel supply design basis for gas-fired edge generation during islanded operation. PE-2 and PE-3 establish the 72-hour minimum fuel reserve as the binding constraint for islanded operation duration. For gas-fired edge generation assets, the fuel gas supply design basis under islanded conditions — when the facility is disconnected from the distribution gas supply infrastructure — requires detailed analysis of the local gas supply architecture. If the microgrid’s gas-fired generation draws fuel from the same distribution system that may be unavailable during islanded operation, an on-site compressed gas or LNG backup supply must be specified.	Gas supply design basis for gas-fired edge generation under islanded conditions established in v0.2 or site-specific implementation guide.
DI-4	Service area defense-critical and critical load inventory. XC-1 requires identification of defense-critical loads within the FC5 entity’s islanding boundary and their mission requirement duration. XC-2 requires identification of critical services (hospitals, water treatment, emergency communications) dependent on the microgrid’s resilience function. This inventory is facility-specific and must be performed by the facility operator for each FC5 installation. Guidance on the inventory methodology is to be developed in a companion implementation guide.	Critical load inventory completed by each FC5 facility operator as part of site-specific design basis implementation.
DI-5	Command Broker communications design basis. The MT-1 and MT-3 Command Broker authorization requirement specifies that the Command Broker must be reachable via a communications channel independent of the primary SCADA path. The specific design basis for this independent channel — technology, bandwidth, reliability requirement, geographic independence from the primary SCADA path — is not yet specified at the series level. This specification is needed to make the Command Broker requirement enforceable as a design basis criterion rather than a procedural aspiration.	Command Broker independent communications channel design basis specified in v0.2 or series-level implementation standard.
DI-6	Dual-DB AI Governance interaction points. Section 4.6 identifies two primary AI governance interaction points for FC5: microgrid controller ML optimization and Gen AI operational decision support. The specific Bright Line boundary for each application — defining which functions are deterministic ML (below Bright Line, autonomous permitted) and which require Command Broker oversight (above Bright Line) — requires detailed analysis of the microgrid control architecture. This analysis is deferred to v0.2.	Bright Line boundary drawn for FC5 microgrid control AI applications; Command Broker requirements specified for Gen AI decision support functions.

Section 7 — References

Regulatory and Standards References

Institute of Electrical and Electronics Engineers. IEEE 1547-2018: Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces. New York: IEEE, 2018.

North American Electric Reliability Corporation. PRC-024-3: Generator Frequency and Voltage Protective Relay Settings. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

Federal Energy Regulatory Commission. Order No. 2222: Participation of Distributed Energy Resource Aggregations in Markets Operated by Regional Transmission Organizations and Independent System Operators. Washington, D.C.: FERC, 2020.

Government and Policy References

Defense Science Board Task Force on DoD Energy Strategy. “More Fight — Less Fuel.” Washington, D.C.: Office of the Under Secretary of Defense for Acquisition, Technology and Logistics, February 2008.

Department of Energy. “QRI Assessment: Quadrennial Energy Review.” Washington, D.C.: DOE, 2015. (Grid resilience and distributed energy resources.)

Technical References

Lasseter, R.H. “MicroGrids.” Proceedings of the 2002 IEEE Power Engineering Society Winter Meeting. New York: IEEE, 2002. (Foundational reference for microgrid islanding architecture.)

Kroposki, B. et al. “Achieving a 100% Renewable Grid: Operating Electric Power Systems with Extremely High Levels of Variable Renewable Energy.” IEEE Power and Energy Magazine, March/April 2017. (Frequency response of inverter-based islanded systems.)

Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES FC5 and CntlCo Gap Analysis. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Predecessor document.)

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GC-FC4: Natural Gas Compressor Station. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. Command Broker Implementation Guide. Version 0.1 — DRAFT. Eclectic Technologies, 2026. (Bright Line and Command Broker architecture reference.)